QP CODE: 24900262

Reg No:.....

Name:.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024

First Semester

Discipline Specific Core Course - MG1DSCTXL100 - TEXTILE SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

[Learning Domain][CO No(s)]

Part A

Multiple Choice Questions

Answer all questions. / each questions carries 1 mark.

1 What happens when cotton fibres are immersed in water? [U] [2]

a) They repel water b) They absorb water

c) They dissolves d) They disintegrates

2 Identify the distinguishing features of silk fibers under microscopic examination. [U] [2]

a) Scales b) Node-like structures

c) Cross-markings d) Fibrils

3 What is a key feature of polyster in terms of durability? [U] [2]

a) it is easily damaged b) it has low tensile strength

c) it is highly durable and strong. d) it deteriorates quickly

4 Identify natural plant-based fiber? [U] [1]

a) wool b) polyester

c) silk d) cotton

5 What is the first step in the processing of cotton after harvesting? [U] [2]

a) carding b) ginning

c) spinning d) weaving

6 Identify the smell while wool burns. [U] [2]

a) Burning of plastic. b) Burning of paper.

c) Burning of feathers or hair. d) Burning of wood.

7 List out Acrylic fiber commonly used for? [U] [2]

a) Outdoor clothing b) Knit wear and Active Wear

c) Upholstery and Furniture d) All of the above

8 Which of the following is a variation of Plain Weave? [U] [4]

a) Twill weave b) Satin weave

c) Rib weave d) Herringbone weave

9 Name the fabric that is made using Jacquard weave. [U] [4]

a) Brocade b) Denim

c) Velvet d) Corduroy

10 Which weave is characterised by a crinkled textured surface? [U] [4]

a) Satin Weave b) Crepe Weave

c) Plain Weave d) Twill Weave

(10 × 1 = 10)

Part B

Fill In The Blanks

Answer all questions

11 The formation of small balls on the surface of fabric is termed as …..... [U] [2]

12 A distinguishing feature of acetate fibre is their …….. [U] [2]
appearance when viewed under a microscope.

13	The process of removing seeds fron cotton fibers called_____.	[U]	[2]
14	Nylon is a ….....fibre.	[U]	[2]
15	Nylon 6 is made from a single six carbon substance called …..... .	[U]	[2]
16	The rotor speed in rotary spinning affects …..... Quality.	[U]	[3]
17	A higher Yarn count indicates a ….....Yarn.	[U]	[3]
18	Bird's eye weave comes under the _____ weave category .	[U]	[4]
19	Lappet weave uses an extra _____ yarn to create patterns .	[U]	[4]
20	All laces are characterised by an _____ structure and decorative design.	[U]	[4]

(10 × 1 = 10)

Part C

Answer In One Word Or One Sentence.

Answer all questions

21	Which property of cotton fibre allows it to absorb moisture?	[U]	[2]
22	What is a common drawback of acrylic fabrics?	[U]	[2]
23	Give one major drawback of synthetic fibers?	[U]	[1]
24	What is a Acetate rayon? and note down the properties.	[U]	[2]
25	which fibre is used for making sports wear?	[U]	[2]
26	What is the Primary Purpose of Ring spinning?	[U]	[3]
27	How is yarn count measured?	[U]	[3]
28	What is the primary purpose of using extra warp yarns in weaving?	[U]	[4]
29	Name another name for the Leno weave.	[U]	[4]
30	Classify twill weave based on the angle of the diagonals.	[U]	[4]

(10 × 1 = 10)

Part D

Short Answer Questions

Answer any 5 questions./ each question carries 2 marks.

31	List down any six properties of wool.	[U]	[2]
32	Why are synthetic fibers preferred in industrial applications?	[U]	[1]
33	What is the significance of combing in cotton processing?	[U]	[2]
34	Define Sericulture.	[U]	[2]
35	Differentiate Single yarn and Ply yarn?	[U]	[3]
36	Illustrate the point paper diagram of a 3x1 right hand twill.	[U]	[4]

(5 × 2 = 10)

Part E

Long Answer Questions.

Answer any two questions./ each question carries 5 marks.

37	Explain the key properties of acetate rayon and discuss its advantages and disadvantages in textile applications.	[U]	[2]
38	Write in detail about Hand spinning?	[U]	[3]
39	Discuss the following - swivel weave, lappet weave, spot weave.	[U]	[4]
40	Explain the physical, chemical and microscopic test of silk.	[U]	[2]

(2 × 5 = 10)

END OF THE QUESTION PAPER

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